

YUJUN (EUGENE) CHEN

Computational Neuroscience · Neural Engineering
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Education & Certifications

Ph.D. Candidate, Biomedical Engineering - University of Florida

Expected December 2026 · Advisor: Dr. Mingzhou Ding

Graduate Certificate, Biomedical Neuroscience - University of Florida

2020

M.S., Information Systems and Operations Management - University of Florida

2013-2015

B.S., Information Systems - South China Agricultural University, Guangzhou, China

2008-2012

Certifications: SAS Certified Advanced Programmer; Oracle Certified Professional (OCP); Red Hat Certified Engineer (RHCE)

Research Experience

Graduate Researcher, Ding Laboratory - University of Florida, J. Crayton Pruitt Family Dept. of Biomedical Engineering

Gainesville, FL · 2022-Present

Experiment design and multimodal data acquisition

- Designed and ran a jittered event-related fMRI study of natural versus diffusion model-generated affective images in 30 participants, using a 1.8 mm CMRR multiband EPI protocol on a 3T scanner with concurrent EyeLink 1000 eye tracking.
- Collected 128-channel EEG, pupillometry, and affect ratings from 109 participants across two versions of an AI-generated affective image study; revised display parameters to remove a pupil light-reflex confound.
- Ran simultaneous EEG-fMRI sessions in 20 participants viewing natural affective scenes, applying gradient and ballistocardiogram artifact correction to preserve trial-by-trial correspondence between modalities.

Neural population analysis and representational dynamics

- Characterized spatial and temporal affective representations using whole-brain fMRI searchlight RSA and EEG-fMRI fusion; analyzed 250 Hz EEG time series and tested onset differences with bootstrap resampling and Kolmogorov-Smirnov tests.
- Compared multimodal LLM and human valence-arousal ratings using representational geometry; measured spatial overlap between their associated fMRI cortical maps to assess brain-model alignment.
- Modeled neural manifolds in image-evoked responses using dimensionality reduction and latent-variable models; tested whether affective and scene-semantic information occupied distinct latent axes.
- Built and benchmarked linear SVM, EEGNet, LSTM, and Transformer decoders for single-trial EEG emotion classification; assessed time-resolved accuracy and decoding gains in the late positive potential window.
- Integrated wilder-attention datasets from independent sites and scanner platforms using Liptak-Stouffer meta-analysis to test replication across cohorts.
- Estimated directed effective connectivity between anterior emotion-modulating regions and visual cortex, then tested whether connectivity strength predicted regional decoding accuracy.
- Tested visual adaptation across repeated images using univariate repetition suppression and sliding-window multivariate decoding to distinguish response fatigue from representational sharpening in primary visual cortex.
- Improved fMRI emotion-category decoding for natural-scene images through TRIBE v2 data augmentation: generated synthetic fMRI responses to unscanned affective images in zero-shot mode, optimized the real-to-synthetic training ratio, and validated gains on held-out in-house and public fMRI data.

Encoding models and brain-inspired AI

- Built an fMRI encoding model using spatial features from vision-language models, ridge regression, a shared latent space, and subject-specific readout heads to address variation across subjects and studies. Despite predominantly neutral training data, predicted responses supported emotion decoding for faces but not affective scenes.
- Used representational similarity analysis to compare human brain responses with CNNs (VGG, ResNet, CORnet), CLIP, and vision-language models, testing how social-emotional information is represented along the mid-level visual pathway.

- Implemented a CLIP-ViT-based Visual Cortex-Amygdala network; used ablation against feedforward controls to isolate reentrant feedback effects. Predicted valence and arousal for affective image datasets ($r = .79$ and $.26$ versus human ratings), with weaker arousal agreement linked to participant-level variability.
- Benchmarked machine-vision saliency models (DeepGaze, TranSalNet, SalFBNet) against human eye-tracking fixations, then examined their generalization from spatial to emotional salience using gradient- and occlusion-based attribution maps.

Analysis pipelines and iterative experiment design

- Built and validated a single-trial GLM pipeline for 1.8 mm fMRI, combining ridge regression, data-driven denoising, and adaptive HRF estimation to improve low-signal trial estimates and support group-level decoding in a common space.
- Integrated EEG, behavioral, and fMRI findings from AI-generated image studies to design and launch a simultaneous EEG-fMRI study with AI-generated affective videos.
- Built documented, version-controlled pipelines for EEG, fMRI, eye tracking, and stimuli that other lab members reuse; shared methods and findings with collaborators in psychology, engineering, and machine learning.

Data Management Analyst III - University of Florida, Clinical and Translational Science Institute

Gainesville, FL · 2020-2022

- Managed the OneFlorida data platform and its i2b2 interface and oversaw UF Health EHR ETL in Python and T-SQL under HIPAA-compliant data operations.
- Transformed UF Clinical Data Warehouse data into the OMOP common data model, the standard schema for longitudinal EHR and observational research, and built reporting workflows in SSIS, SSRS, SSAS, and data visualization tools like Tableau.

Data Management Analyst II - University of Florida, Office of Institutional Planning & Research

Gainesville, FL · 2016-2020

- Led the analytics behind a ranking-improvement program supporting UF's move from No. 14 to No. 6 among U.S. News public universities, using what-if, time-series, and causal analyses to identify actionable policy levers.

Selected Publications and Presentations

Chen, Y., Bo, K., Keil, A., & Ding, M. (submitted). Joint representation of valence and arousal in the human brain.

Chen, Y., Gilbert, F., Fang, R., Keil, A., & Ding, M. (submitted). Neural responses to natural versus AI-generated affective images.

Bo, K., Cui, L., Chen, Y., Chen, G., Keil, A., & Ding, M. (2026). Neural representation of emotional valence in human amygdala and surrounding regions. *NeuroImage*, 326, 121693.

Liu, P., Bo, K., Chen, Y., Keil, A., Ding, M., & Fang, R. (2026). Biologically inspired deep neural network models for visual emotion processing. *Network Neuroscience*. Advance online publication.

Xiong, C., Chen, Y., Yang, Q., Kim, S., Meyyappan, S., Bengson, J. J., Mangun, G. R., & Ding, M. (2026). Neural mechanisms of willed attention control [Reviewed preprint]. *eLife*, 15, RP110199.

Cui, L., Bo, K., Xiong, C., Chen, Y., Keil, A., & Ding, M. (2025). Stimulus repetition induces a two-stage learning process in the primary visual cortex. *Journal of Neuroscience*, 45(23), e1788242025.

Gilbert, F., Yang, F., Smith, E., Gardy, S. M., Farkas, A. H., Chen, Y., Ding, M., Fang, R., & Keil, A. (2025). Emotional responses to naturalistic and AI-generated affective pictures: A systematic comparison. *bioRxiv*.

Chen, Y., Cui, L., & Ding, M. (2023). Neural processing of affective scenes: A comparison between convolutional neural networks and human visual pathways. *Journal of Vision*, 23(9), 5044.

Technical Skills

Neural data modeling: GLM; encoding and decoding models; MVPA; RSA; connectivity analysis; dimensionality reduction and latent-variable models; state-space and neural manifold analysis

Neuroimaging tools: fMRIPrep, SPM, FreeSurfer, FSL, GLMsingle, Nilearn, MRICroGL, pycortex, BIDS

EEG processing and analysis: BrainVision Analyzer; EEGLAB and EMEGS for filtering, epoching, re-referencing, and SOBI/ICA correction; Psychtoolbox; ERP and single-trial decoding; weight maps; time-frequency and temporal-generalization maps

Machine learning: SVM, CNN, LSTM, Transformer, and CLIP-ViT architectures; supervised training and fine-tuning of pretrained models; cross-validation and held-out evaluation; hyperparameter tuning and regularization; ablation and benchmarking across architectures; brain-model alignment; feedback/reentry models; saliency and attribution methods

Statistics and inference: Permutation and cluster-based testing; bootstrap resampling and uncertainty estimation; FDR and FWE correction; variance partitioning; cross-validated generalization testing

Languages and libraries: Python (NumPy, SciPy, scikit-learn, PyTorch, Nilearn); MATLAB (EEGLAB, SPM, Psychtoolbox); SQL

Infrastructure: Git, Docker, Linux/WSL, AWS, Hugging Face, Slurm and HiPerGator GPU cluster computing, open data release, JAX